Health-a-thon 2026

FINAL SUBMISSION • CANCER CARE TRACK

OncoBrief

The consult starts before the patient walks in: source-linked record readiness, smart OPD flow, and caregiver voice continuity.

EVIDENCE LEDGER
Every fact links to page, span & bounding box

RECONCILIATION ROOM
Conflicting dates shown side by side for doctors

SMART OPD & STUDIO
Readiness-gated slot backfill & 4 cited drafts

11-LOCALE CONTINUITY
Consented caregiver sync & Sarvam Bulbul v3

TEAM NAME: ICONIX (R.M.K. Engineering College)

TRACK & USE CASES: Cancer Care (UC 02, 04, 05, 01)

Team Overview & Background

Built by R.M.K. Engineering College technologists alongside an Oncology Clinical Lead for the National Cancer Grid ecosystem.

ENGINEERING TEAM • ICONIX



- **Dhaanish Ahamed & ICONIX Team:** R.M.K. Engineering College, Chennai
- **Full-Stack & Ledger Architecture:** Built the 6-package TypeScript monorepo, Next.js 15 App Router, and PostgreSQL 16 append-only Evidence Ledger with 42 forced-RLS tables.
- **AI & Cryptographic Pipeline:** Engineered the deterministic span validation gate (validateQuoteInSpans), X25519 ECDH Fidelius decryption, and AWS ap-south-1 Fargate stack.
- **Verification Rigor:** Authored 886 passing unit/integration tests and 51 Chromium E2E tests across 15 SQL migrations.

DOCTOR PARTNER & CLINICAL LEAD



- **Clinical Lead, Oncology:** Practising Oncology Specialist Partner
- **NCG Workflow & Checklist Design:** Authored the 3 oncology document readiness templates: NCG Referral Intake, Multidisciplinary Tumour Board Prep, and Oncology Follow-Up.
- **Strict Non-Diagnostic Boundary:** Defined the 17 forbidden clinical concepts enforced in code: zero staging inference, zero treatment advice, and zero risk scoring.
- **Human-in-the-Loop Governance:** Designed the side-by-side Reconciliation Room so conflicting surgical or lab dates require clinician sign-off.

INSTITUTION & ECOSYSTEM FIT



- **R.M.K. Engineering College:** Department of Computer Science & Engineering
- **NCG & Koita Centre Alignment:** Designed around the National Cancer Grid (370+ hospitals treating 60% of India cancer cases) and IIT Bombay KCDH operational standards.
- **Built for Real Indian OPDs:** Operates on plastic-bag paper referrals, scanned lab PDFs, re-faxed reports, and low-literacy family caregiver workflows across 11 locales.
- **Zero-EMR Disruption:** Runs as a pre-EMR readiness layer alongside hospital HIS systems without requiring custom database write-back.



HOW ENGINEERING AND ONCOLOGY CO-DESIGNED ONCOBRIEF V4

1. Clinical Problem Discovery: Mapped why consultations stall when patients bring records from 4 to 6 outside facilities, and why serial CBC blood counts must form dated flowsheets instead of false contradictions.

2. Multi-Role Expansion: Expanded the initial intake prototype to cover Consultation Readiness (UC 02), Smart OPD Queue & Waitlist Backfill (UC 04), Grounded Clinician Studio (UC 05), and 11-Locale Caregiver Voice Continuity (UC 01).

Problem Statement and Supporting Evidence

LIMITATION IN EXISTING TOOLS

- Summaries show what is available, not what was planned, completed, or missed.
- Referral trackers show referrals, but not whether the required report returned or is current.
- Readiness is not tracked, so departments cannot see why consultation could not proceed.
- Serial lab counts get misread as conflicts, and OPD slots stall on missing reports.
- No unified ledger connects source spans, readiness checklists, OPD slots, and caregivers.



Next visit

is when reports typically become available, in a large Indian cancer centre's own description of its OPD



6 sources

surgery, pathology, imaging, chemotherapy, radiation and outside centres feed one oncology record



14% to 30%

of Indian breast cancer patients are lost to follow-up by five years



Target Users and Use Case

Four oncology care roles working off one append-only Evidence Ledger across tertiary cancer centres and district spokes.

1. ONCOLOGISTS (MEDICAL, SURGICAL, RADIATION)

Use Case 02 & Use Case 05 • Tertiary OPDs & Tumour Boards



- **Healthcare Setting:** High-volume oncology outpatient clinics (30 to 50 patients/day) and multidisciplinary tumour boards.
- **Core Workflow Challenge:** Rebuilding a timeline from 6+ outside reports takes consultation time away from the patient.
- **How OncoBrief Fits:** Opens with the "Changes Since Last Consultation" banner, dated serial lab flowsheets, 2-click batch verification, side-by-side Reconciliation Room, and 4 source-cited Studio drafts.

2. OPD & TUMOUR BOARD COORDINATORS

Use Case 04 • Clinic Front Desks & Coordination Rooms



- **Healthcare Setting:** Outpatient waiting areas, scheduling counters, and tumour board prep teams.
- **Core Workflow Challenge:** Cancelled slots or delayed investigations leave specialist rooms idle while ready patients wait.
- **How OncoBrief Fits:** Provides a live OPD Slot & Queue Board, identifies record-ready waitlist backfill candidates in 9.49 ms, and assigns document retrieval tasks to named staff with deadlines.

3. RECORDS OFFICERS & INTAKE STAFF

Use Case 02 • Medical Records & Referral Counters



- **Healthcare Setting:** Hospital intake desks receiving paper files, lab PDFs, and external hospital referrals.
- **Core Workflow Challenge:** Duplicate faxes, unindexed scans, and missing histopathology or IHC reports slip into clinic.
- **How OncoBrief Fits:** Ingests PDFs, mobile photos, HL7 mCODE STU4, and ABDM v6.5.0 bundles; flags near-duplicate faxes; and tracks NCG-aligned intake checklists.

4. CANCER PATIENTS & FAMILY CAREGIVERS

Use Case 01 • Home Preparation & Inter-City Travel



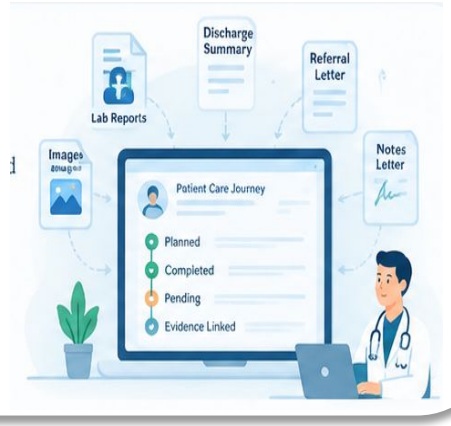
- **Healthcare Setting:** Patients and consented family caregivers travelling across states for multi-cycle therapy.
- **Core Workflow Challenge:** English discharge notes and missed pre-visit instructions cause avoidable travel delays and drop-offs.
- **How OncoBrief Fits:** Synchronizes clinician-approved reminders to consented caregivers with 11-locale voice note previews (Sarvam Bulbul v3 / browser fallback) and two-way attendance logging.

Solution Overview & Key Features (1/2)

One ledger of source-linked facts. The timeline is only a view over it.

THE SOLUTION

- OncoBrief ingests fragmented patient documents from multiple sources, analyses them with AI, and converts them into a structured, evidence-linked care journey
- It gives clinicians a complete, up-to-date view of what was planned, what happened, what is missing, and who is responsible for the next action.



NOVELTY

Evidence Ledger - Every fact is linked to its original source, sentence, and page.

2-Click Batch Verification & Delta Brief: Highlights changes since last visit, serial lab flowsheets, and batch-verifies >=90% OCR facts.

Reconciliation Room - Conflicting documents are shown side by side for review and resolution.

Smart OPD, Studio & 11-Locale Caregivers: Readiness-gated OPD slot backfill, 4 cited Studio drafts, and Sarvam Bulbul v3 voice notes.

READS

Scans and understands fragmented records such as lab reports, discharge summaries, referral letters, photographs and handwritten notes.





Solution Overview: User Journey & Impact (2/2)

End-to-end journey: Ingest (PDF/ABDM/mCODE) -> Span Gate -> 2-Click Verify & Reconcile -> Smart OPD, Studio & Caregiver Audio.

370+

cancer centres in the National Cancer Grid, treating close to 60% of India's patients

~8.5 lakh

new cancer cases a year pass through those member centres

Zero

of them can currently report what share of their consults were ready to proceed

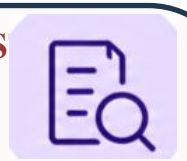
CONTINUITY



Before: Care journeys can become fragmented when patients move between hospitals.

OncoBrief: Preserves the administrative care state so the journey can continue without resetting.

EVIDENCE TRUS



Before: Facts may lack a clear link to their original document and source.

OncoBrief : Maintains **Document → Phrase → Fact → Task** provenance for traceable and verifiable evidence.

COORDINATION



Before: Tasks can have unclear ownership and ambiguous closure.

OncoBrief : Assigns responsibility, tracks accountability, and records **human-confirmed closure**.

OPERATIONAL EFFICIENCY

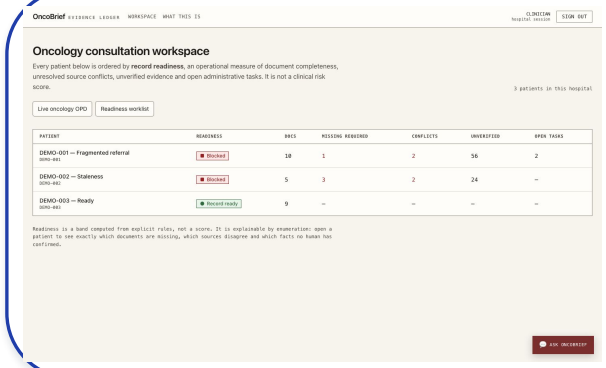


Before: Missing, duplicate, or conflicting documents can cause delays and rework.

OncoBrief : Identifies evidence gaps, duplicates, and conflicts so coordinators can reconcile them before closure.

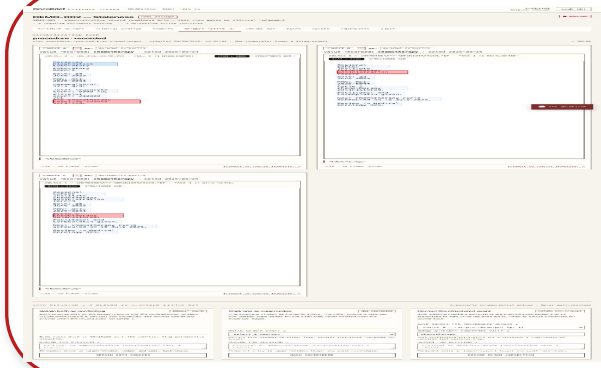
Solution Working MVP / Demo

LIVE AWS MUMBAI MVP: **https://dsc1vmsr4q09g.cloudfront.net** | **GITHUB:** **github.com/dhaanish597/OncoBrief** | **DEMO LOGINS:** dr.rao@rci.demo • coord.anita@rci.demo • auditor.k@rci.demo



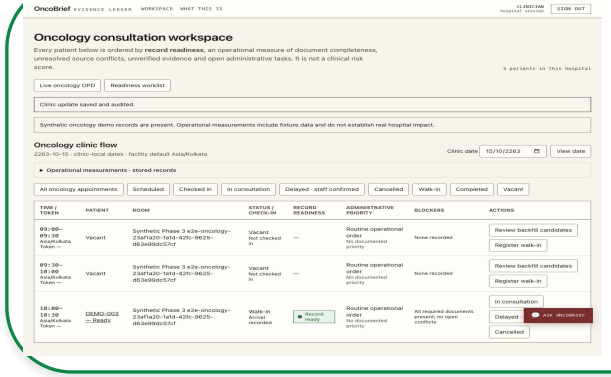
1. Consultation Readiness Worklist (/workspace)

Orders oncology referrals by deterministic Readiness Band (DEMO-001 & DEMO-002 Blocked, DEMO-003 Record Ready) with missing-document and conflict counts.



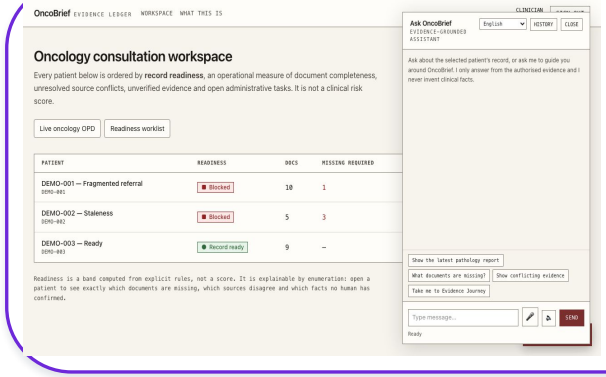
2. Side-by-Side Reconciliation Room (/conflicts)

Displays competing source PDFs with highlighted bounding-box spans (12 Jun 2025 vs. 3 Jul 2025 mastectomy) for clinician sign-off.



3. Smart Oncology OPD Slot & Backfill Board

Tracks scheduled, checked-in, delayed-lab, and vacant slots; ranks record-ready waitlisted patients in 9.49 ms for human-confirmed backfill.



4. Grounded Co-Pilot, Studio & 11-Locale Audio

Retrieves [E#] patient spans & [G#] SOP references, compiles 4 cited Studio drafts (SHA-256 frozen), and previews 11-locale caregiver voice notes.

DEMO-001: Fragmented Referral (Blocked)

10 docs across 4 centres, duplicate CT chest fax, serial Hb delta (-1.2 g/dL), 2 open conflicts, 1 missing insurance auth.

DEMO-002: Staleness Re-Opening (Blocked)

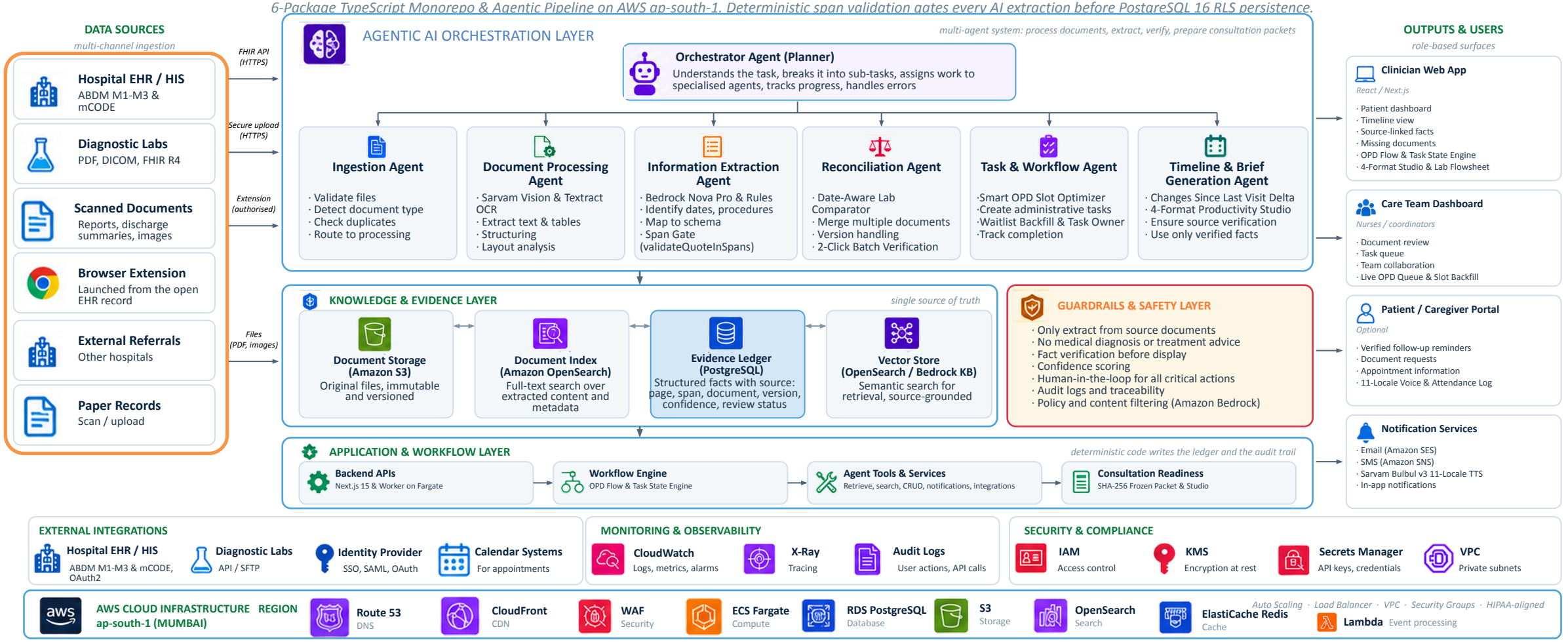
5 docs showing how a new contradictory report re-opens a verified appointment fact and marks an earlier packet stale.

DEMO-003: Verified & Ready (Green)

9 verified docs, 0 conflicts: ready for packet freeze, 4-format Studio export (ABDM/mCODE), OPD walk-in, and caregiver audio.

Technology, Tools & V4 System Architecture

6-Package TypeScript Monorepo & Agentic Pipeline on AWS ap-south-1. Deterministic span validation gates every AI extraction before PostgreSQL 16 RLS persistence.



Build Sprint Journey and Progress

Four-week build sprint evolving OncoBrief from a new-patient intake ledger into a 4-use-case oncology operations platform.



WEEK 1 • FOUNDATION
Core Evidence Ledger & AWS Stack

- Built 6-package TypeScript monorepo & PostgreSQL 16 schema.
- Enforced append-only ledger_entry & audit_event SHA-256 chains.
- Wired deterministic span gate (validateQuoteInSpans).
- Deployed live stack on AWS Mumbai (Fargate, RDS, S3, Textract, Bedrock).

WEEK 2 • READINESS (UC 02)
Longitudinal Deltas & Reconciliation

- Built "Changes Since Last Consultation" interval delta banner.
- Added date-aware serial CBC/lab flowsheets (-1.2 g/dL delta).
- Added 2-click atomic batch verification (>=90% OCR filter).
- Fixed Reconciliation Room state machine to supersede or correct conflicts.

WEEK 3 • OPD & STUDIO (UC 04/05)
Smart Slot Backfill & 4 Drafts

- Built OPD slot state machine & 9.49 ms waitlist backfill engine.
- Added 1-click task ownership & coordinator assignment controls.
- Added [G#] institutional SOP lookup with bounding-box preview.
- Built 4-format Clinician Studio with 64-char SHA-256 freeze.

WEEK 4 • CONTINUITY & AUDIT
11-Locale Audio & Security Locks

- Built consented caregiver registry & immediate revocation lock.
- Added 11-locale audio player (Sarvam Bulbul v3 + Web Speech fallback).
- Added two-way attendance logging & 15-key PHI audit redaction.
- Passed 886 unit/integration & 51 Chromium E2E acceptance tests.

MAJOR CHALLENGES ENCOUNTERED & RESOLVED

- **False Conflicts on Serial Labs:** Scoped comparators by observed_on date so June vs. July CBCs form a flowsheet while same-date disagreements open a conflict.
- **Consent Withdrawal vs. Immutable Ledger:** Applied RESTRICTIVE RLS policies to mask withdrawn ABDM sources and block exports (409) without breaking historical SHA-256 chains.

PLANNED CAPABILITIES REMAINING FOR PILOT

- **EHR Browser Extension Overlay:** Permission-scoped browser launch layer (apps/extension) is planned for phase-2 hospital rollout.
- **Live Carrier & NHA Sandbox Gateways:** SMS/WhatsApp delivery and ABDM M3 callbacks run via badged local simulation until TRAI DLT and NHA production keys are issued.

Evaluation and User Validation

886 Passed

Automated unit & integration tests (0 failed) across domain, db, adapters, worker & web

51 E2E Passed

Chromium Playwright browser tests across 12 specs including 10-scene judge journey (4.81s)

45+ -> 2 Clicks

Clinician chart verification effort reduced via >=90% OCR atomic batch verification

9.49 ms

Readiness-gated OPD waitlist backfill search latency (0.88 ms on occupied-slot fast path)

AUTOMATED VERIFICATION & MODEL GROUNDING RESULTS



- **Package Breakdown (pnpm verify):** domain (415), db (183), web (173), adapters (86), worker (29) = 886 passed, 0 failed.
- **100% Span Entailment Gate:** 0 unanchored AI facts enter PostgreSQL; validateQuoteInSpans rejects any hallucinated quote or altered lab unit.
- **Cryptographic Hash Chain Audit:** 159 ledger entries and 147 audit events independently recomputed from genesis with 100% SHA-256 payload match.
- **Security & RLS Regressions:** Verified cross-tenant 404 isolation across 51 RLS tables, Fidelius X25519 key wiping, and 409 lock after source revocation.

USABILITY TESTING & CLINICAL WORKFLOW IMPROVEMENTS



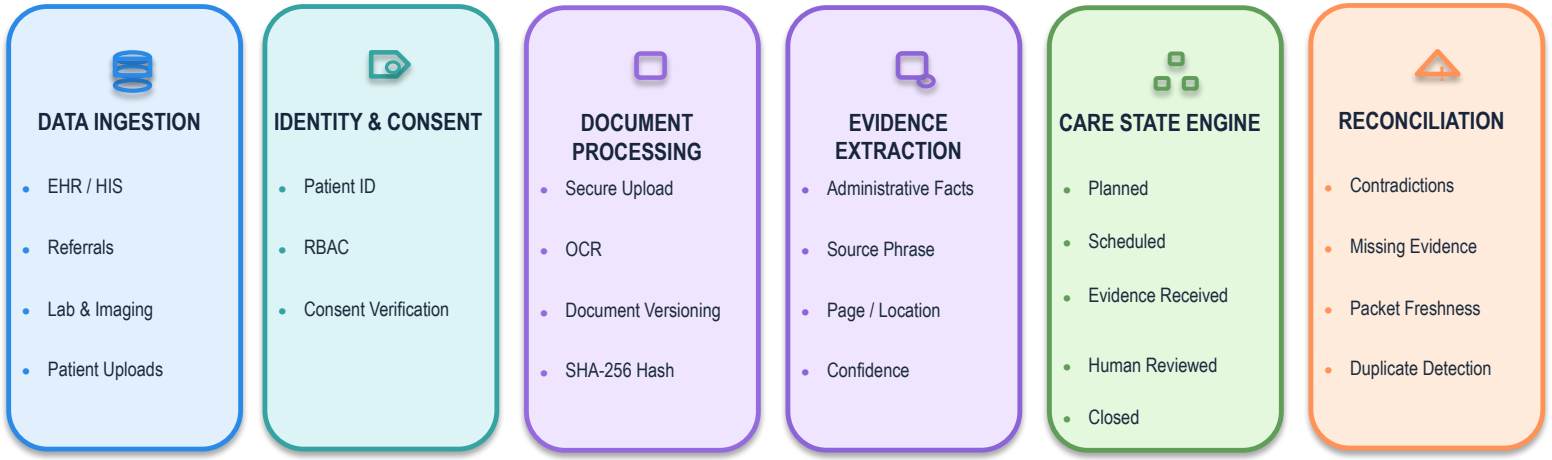
- **Reconciliation Room Deadlock Fixed:** Added provenance base selector and human replacement value input directly in Outcome 3 so clinicians can resolve or supersede conflicts in one screen.
- **Direct Page & Bbox Source Jump:** Upgraded serial lab flowsheet and medication links to route directly to /sources/[docId]?fact=[id]&page=[p] with A4 highlight geometry.
- **WCAG 2.2 AA & 375px Mobile Polish:** Remediated contrast tokens (6.12:1 extracted, 5.10:1 conflict) and added pb-24 clearance across 375px, 768px, 1024px, and 1440px viewports.

EPISTEMIC HONESTY: LOCALLY VERIFIED EXECUTION VS. AREAS REQUIRING HOSPITAL PILOT VALIDATION

- **Working Locally Today:** PostgreSQL 16 RLS, SHA-256 hash chains, local PDF/FHIR/DICOM/vCard parsing, Fidelius X25519 ECDH decryption, OPD optimizer, and Studio generators.
- **Simulated / Pilot Validation Needed:** SMS/WhatsApp outbox and ABDM gateway callbacks are simulated locally; spoken regional TTS requires Sarvam API keys or installed OS voices; real-world consultation minutes saved will be measured in an ethics-approved hospital shadow pilot.

Safety and Responsible AI

Deterministic code owns state transitions, RLS privacy, and consent locks; AI models only propose span-checked candidates.



17 FORBIDDEN CLINICAL CONCEPTS ENFORCED BY SCHEMA & CI SCANNER (clinical-boundary.ts)

Diagnosis & staging
no diagnosis, stage or severity field exists

Treatment advice
only repeats what a named clinician wrote

Decision support
reports what was asked, never what to do

Risk scoring
OPD queue uses readiness & admin tags only

Data interpretation
lab flowsheets show arithmetic deltas only

Autonomous action
409 lock on revoked sources; 15-key PHI redaction

Deployment, Cost and Sustainability Plan

How OncoBrief reaches 370 cancer centres and keeps running after the hackathon.

COST STRUCTURE

- One-time setup: ₹40k to ₹75k/facility. Per 10-page packet: ₹1.50 to ₹4.00 (₹0 on local PDF fallback).
- *Hosting:* one container host, near zero at pilot scale.



REVENUE STREAMS

- Annual per-centre SaaS licence tiered by OPD volume (spokes vs. tertiary cancer hubs).
- Cross-subsidized public spokes via CSR & NCG-Koita digital oncology grants.



SUSTAINABILITY

Fewer wasted journeys for the patients who travel furthest — no new hardware, it runs on the paper already in the room — an open-source core keeps the cost per document near zero — one engine serves diabetes and maternal health by swapping a checklist file.

What Next? Plans to Take Initiative Forward

A concrete 90-day post-hackathon roadmap from observational shadow pilot to National Cancer Grid ecosystem rollout.

PHASE 1 • DAYS 1 TO 30
Observational Shadow Pilot & Baseline

- **Hospital Shadow Pilot:** Deploy alongside existing EMRs in a partner oncology OPD under institutional ethics approval using de-identified and consented records.
- **Baseline Telemetry:** Measure real-world chart prep time, % of consults starting with complete checklists, and contradiction catch rates.
- **Clinician Co-Design:** Refine NCG intake, tumour board, and follow-up checklist packs with surgical, medical, and radiation oncology leads.

PHASE 2 • DAYS 31 TO 60
Gateway Certification & Edge Hardening

- **NHA ABDM Certification:** Complete official NHA ABDM Sandbox M1/M2/M3 compliance testing and X.509 production gateway onboarding.
- **Live Telecom & Voice:** Wire TRAI DLT-registered WhatsApp/SMS aggregators and live Sarvam Bulbul v3 / Saaras v3 voice pipelines for regional caregivers.
- **Security Hardening:** Enable KMS key rotation, AWS WAF rules, and off-box cryptographic audit anchoring.

PHASE 3 • DAYS 61 TO 90+
Browser Extension & Multi-Centre Scale

- **EHR Browser Launch Layer:** Ship the permission-scoped browser extension (apps/extension) for 1-click chart context handoff without silent scraping.
- **NCG Spoke Expansion:** Scale across NCG tertiary hubs and district cancer spokes via hub-and-spoke SaaS licensing.
- **Grant & Funding Pathway:** Apply for Koita Foundation, BIRAC, and NCG-KCDO digital oncology deployment grants.



PUBLIC AVAILABILITY CONFIRMATION (TEMPLATE QUESTION)

Question: *Would you like to make all the solution information publicly available (e.g., shared on Health-a-thon or other websites)?*

Answer: YES. We are glad to share our solution overview, non-diagnostic Evidence Ledger architecture, and open technical documentation publicly on Health-a-thon and partner platforms to support safe oncology operations across India.

Health-a-thon 2026

Thank you

TEAM ICONIX (R.M.K. Engineering College) • CANCER CARE TRACK

clashanimations597@gmail.com | MVP: https://dsc1vmsr4q09g.cloudfront.net | GitHub: github.com/dhaanish597/OncoBrief

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17 Boundary Guards
Zero diagnosis or risk scoring

AWS Mumbai Live
ECS Fargate, RDS, S3, Bedrock